Concepts

Astra Trident

NetApp
February 12, 2024

This PDF was generated from https://docs.netapp.com/us-en/trident-2304/trident-concepts/intro.html on February 12, 2024. Always check docs.netapp.com for the latest.

Table of Contents

Concepts . . . 1
Learn about Astra Trident . . . 1
ONTAP drivers . . . 2
Provisioning . . . 3
Volume snapshots . . . 4
Virtual pools . . . 4
Volume access groups . . . 6

Concepts

Learn about Astra Trident

Astra Trident is a fully supported open source project maintained by NetApp as part of the Astra product family. It has been designed to help you meet your containerized applications' persistence demands using industry-standard interfaces, such as the Container Storage Interface (CSI).

Overview

Astra Trident deploys in Kubernetes clusters as pods and provides dynamic storage orchestration services for your Kubernetes workloads. It enables your containerized applications to quickly and easily consume persistent storage from NetApp's broad portfolio that includes ONTAP (AFF/FAS/Select/Cloud/Amazon FSx for NetApp ONTAP), Element software (NetApp HCI/SolidFire), as well as the Azure NetApp Files service, and Cloud Volumes Service on Google Cloud.

Astra Trident is also a foundational technology for NetApp's Astra, which addresses your data protection, disaster recovery, portability, and migration use cases for Kubernetes workloads leveraging NetApp's industry-leading data management technology for snapshots, backups, replication, and cloning.

Supported Kubernetes cluster architectures

Astra Trident is supported with the following Kubernetes architectures:

Kubernetes cluster architectures	Supported	Default install
Single master, compute	Yes	Yes
Multiple master, compute	Yes	Yes
Master, `etcd`, compute	Yes	Yes
Master, infrastructure, compute	Yes	Yes

What is Astra?

Astra makes it easier for enterprises to manage, protect, and move their data-rich containerized workloads running on Kubernetes within and across public clouds and on-premises. Astra provisions and provides persistent container storage using Astra Trident from NetApp's proven and expansive storage portfolio in the public cloud and on-premises. It also offers a rich set of advanced application-aware data management functionality, such as snapshot, backup and restore, activity logs, and active cloning for data protection, disaster/data recovery, data audit, and migration use-cases for Kubernetes workloads.

You can sign up for a free trial on the Astra page.

For more information

- NetApp Astra product family

- Astra Control Service documentation
- Astra Control Center documentation
- Astra API documentation

ONTAP drivers

Astra Trident provides five unique ONTAP storage drivers for communicating with ONTAP clusters.

Astra Control supported drivers

Astra Control provides seamless protection, disaster recovery, and mobility (moving volumes between Kubernetes clusters) for volumes created with the `ontap-nas`, `ontap-nas-flexgroup`, and `ontap-san` drivers. See Astra Control replication prerequisites for details.

- You must use `ontap-nas` for production workloads that require data protection, disaster recovery, and mobility.
- Use `ontap-san-economy` when anticipated volume usage is expected to be much higher than what ONTAP supports.
- Use `ontap-nas-economy` only where anticipated volume usage is expected to be much higher than what ONTAP supports, and the `ontap-san-economy` driver cannot be used.
- Do not use use `ontap-nas-economy` if you anticipate the need for data protection, disaster recovery, or mobility.

Astra Trident storage drivers for ONTAP

Astra Trident provides the following storage drivers to communicate with the ONTAP cluster. Supported access modes are: *ReadWriteOnce* (RWO), *ReadOnlyMany* (ROX), *ReadWriteMany* (RWX), *ReadWriteOncePod* (RWOP).

Driver	Protocol	volumeMode	Access modes supported	File systems supported
`ontap-nas`	NFS SMB	Filesystem	RWO, ROX, RWX, RWOP	"", nfs, smb
`ontap-nas-economy`	NFS SMB	Filesystem	RWO, ROX, RWX, RWOP	"", nfs, smb
`ontap-nas-flexgroup`	NFS SMB	Filesystem	RWO, ROX, RWX, RWOP	"", nfs, smb
`ontap-san`	iSCSI	Block	RWO, ROX, RWX, RWOP	No filesystem; raw block device

Driver	Protocol	volumeMode	Access modes supported	File systems supported
`ontap-san`	iSCSI	Filesystem	RWO, ROX, RWOP RWX is not available in Filesystem volume mode.	`xfs`, `ext3`, `ext4`
`ontap-san-economy`	iSCSI	Block	RWO, ROX, RWX, RWOP	No filesystem; raw block device
`ontap-san-economy`	iSCSI	Filesystem	RWO, ROX, RWOP RWX is not available in Filesystem volume mode.	`xfs`, `ext3`, `ext4`

> ONTAP backends can be authenticated by using login credentials for a security role (username/password) or the private key and the certificate that is installed on the ONTAP cluster. You can update existing backends to move from one authentication mode to the other with `tridentctl update backend`.

Provisioning

Provisioning in Astra Trident has two primary phases. The first phase associates a storage class with the set of suitable backend storage pools and occurs as a necessary preparation before provisioning. The second phase includes the volume creation itself and requires choosing a storage pool from those associated with the pending volume's storage class.

Storage class association

Associating backend storage pools with a storage class relies on both the storage class's requested attributes and its `storagePools`, `additionalStoragePools`, and `excludeStoragePools` lists. When you create a storage class, Trident compares the attributes and pools offered by each of its backends to those requested by the storage class. If a storage pool's attributes and name match all of the requested attributes and pool names, Astra Trident adds that storage pool to the set of suitable storage pools for that storage class. In addition, Astra Trident adds all storage pools listed in the `additionalStoragePools` list to that set, even if their attributes do not fulfill all or any of the storage class's requested attributes. You should use the `excludeStoragePools` list to override and remove storage pools from use for a storage class. Astra Trident performs a similar process every time you add a new backend, checking whether its storage pools satisfy those of the existing storage classes and removing any that have been marked as excluded.

Volume creation

Astra Trident then uses the associations between storage classes and storage pools to determine where to provision volumes. When you create a volume, Astra Trident first gets the set of storage pools for that volume's storage class, and, if you specify a protocol for the volume, Astra Trident removes those storage pools that cannot provide the requested protocol (for example, a NetApp HCI/SolidFire backend cannot provide a file-based volume while an ONTAP NAS backend cannot provide a block-based volume). Astra Trident randomizes the order of this resulting set, to facilitate an even distribution of volumes, and then iterates through it,

attempting to provision the volume on each storage pool in turn. If it succeeds on one, it returns successfully, logging any failures encountered in the process. Astra Trident returns a failure **only if** it fails to provision on **all** the storage pools available for the requested storage class and protocol.

Volume snapshots

Learn more about how Astra Trident handles the creation of volume snapshots for its drivers.

Learn about volume snapshot creation

- For the `ontap-nas`, `ontap-san`, `gcp-cvs`, and `azure-netapp-files` drivers, each Persistent Volume (PV) maps to a FlexVol. As a result, volume snapshots are created as NetApp snapshots. NetApp's snapshot technology delivers more stability, scalability, recoverability, and performance than competing snapshot technologies. These snapshot copies are extremely efficient both in the time needed to create them and in storage space.
- For the `ontap-nas-flexgroup` driver, each Persistent Volume (PV) maps to a FlexGroup. As a result, volume snapshots are created as NetApp FlexGroup snapshots. NetApp's snapshot technology delivers more stability, scalability, recoverability, and performance than competing snapshot technologies. These snapshot copies are extremely efficient both in the time needed to create them and in storage space.
- For the `ontap-san-economy` driver, PVs map to LUNs created on shared FlexVols. VolumeSnapshots of PVs are achieved by performing FlexClones of the associated LUN. ONTAP's FlexClone technology makes it possible to create copies of even the largest datasets almost instantaneously. Copies share data blocks with their parents, consuming no storage except what is required for metadata.
- For the `solidfire-san` driver, each PV maps to a LUN created on the NetApp Element software/NetApp HCI cluster. VolumeSnapshots are represented by Element snapshots of the underlying LUN. These snapshots are point-in-time copies and only take up a small amount of system resources and space.
- When working with the `ontap-nas` and `ontap-san` drivers, ONTAP snapshots are point-in-time copies of the FlexVol and consume space on the FlexVol itself. This can result in the amount of writable space in the volume to reduce with time as snapshots are created/scheduled. One simple way of addressing this is to grow the volume by resizing through Kubernetes. Another option is to delete snapshots that are no longer required. When a VolumeSnapshot created through Kubernetes is deleted, Astra Trident will delete the associated ONTAP snapshot. ONTAP snapshots that were not created through Kubernetes can also be deleted.

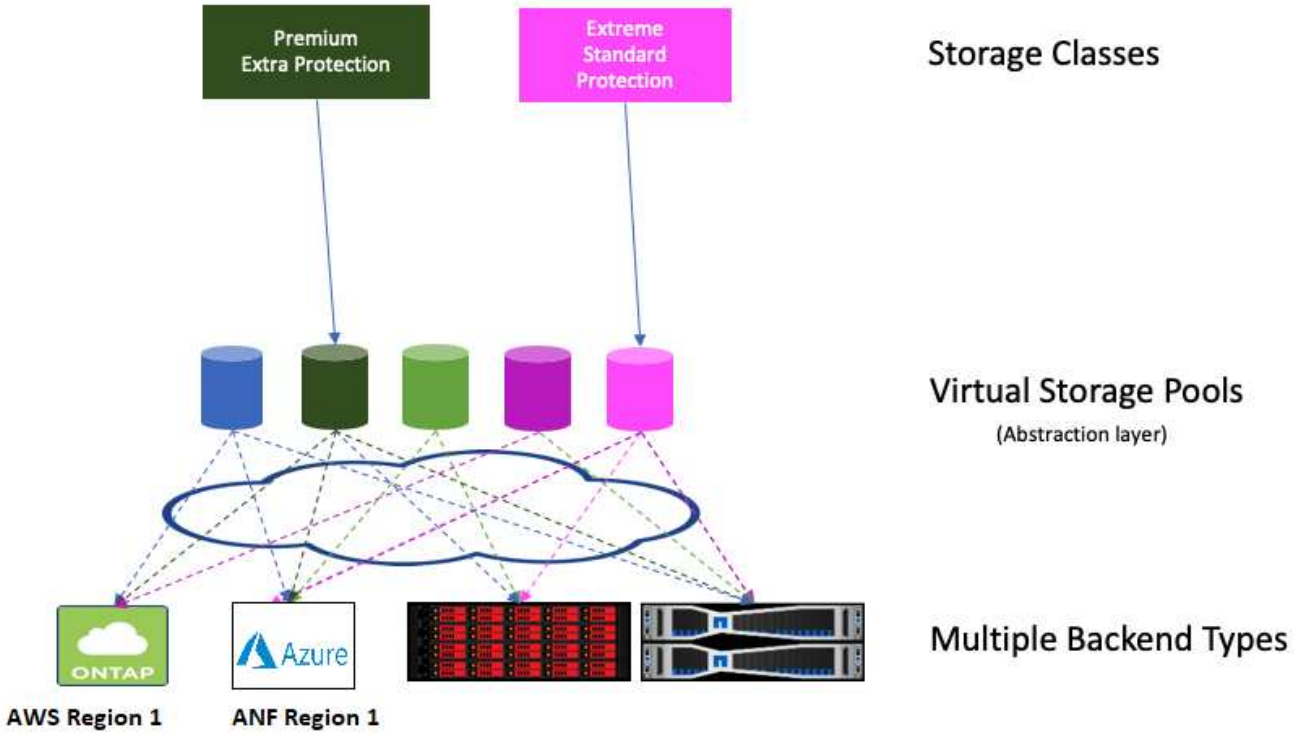
With Astra Trident, you can use VolumeSnapshots to create new PVs from them. Creating PVs from these snapshots is performed by using the FlexClone technology for supported ONTAP and CVS backends. When creating a PV from a snapshot, the backing volume is a FlexClone of the snapshot's parent volume. The `solidfire-san` driver uses Element software volume clones to create PVs from snapshots. Here it creates a clone from the Element snapshot.

Virtual pools

Virtual pools provide a layer of abstraction between Astra Trident storage backends and Kubernetes `StorageClasses`. They allow an administrator to define aspects, such as location, performance, and protection for each backend in a common, backend-agnostic way without making a `StorageClass` specify which physical backend, backend pool, or backend type to use to meet desired criteria.

Learn about virtual pools

The storage administrator can define virtual pools on any of the Astra Trident backends in a JSON or YAML definition file.



Any aspect specified outside the virtual pools list is global to the backend and will apply to all the virtual pools, while each virtual pool might specify one or more aspects individually (overriding any backend-global aspects).

- When defining virtual pools, do not attempt to rearrange the order of existing virtual pools in a backend definition.
- We advise against modifying attributes for an existing virtual pool. You should define a new virtual pool to make changes.

Most aspects are specified in backend-specific terms. Crucially, the aspect values are not exposed outside the backend's driver and are not available for matching in `StorageClasses`. Instead, the administrator defines one or more labels for each virtual pool. Each label is a key:value pair, and labels might be common across unique backends. Like aspects, labels can be specified per-pool or global to the backend. Unlike aspects, which have predefined names and values, the administrator has full discretion to define label keys and values as needed. For convenience, storage administrators can define labels per virtual pool and group volumes by label.

A `StorageClass` identifies which virtual pool to use by referencing the labels within a selector parameter. Virtual pool selectors support the following operators:

Operator	Example	A pool's label value must:
=	performance=premium	Match

Operator	Example	A pool's label value must:
`!=`	performance!=extreme	Not match
`in`	location in (east, west)	Be in the set of values
`notin`	performance notin (silver, bronze)	Not be in the set of values
`<key>`	protection	Exist with any value
`!<key>`	!protection	Not exist

Volume access groups

Learn more about how Astra Trident uses volume access groups.

> Ignore this section if you are using CHAP, which is recommended to simplify management and avoid the scaling limit described below. In addition, if you are using Astra Trident in CSI mode, you can ignore this section. Astra Trident uses CHAP when installed as an enhanced CSI provisioner.

Learn about volume access groups

Astra Trident can use volume access groups to control access to the volumes that it provisions. If CHAP is disabled, it expects to find an access group called `trident` unless you specify one or more access group IDs in the configuration.

While Astra Trident associates new volumes with the configured access group(s), it does not create or otherwise manage access groups themselves. The access group(s) must exist before the storage backend is added to Astra Trident, and they need to contain the iSCSI IQNs from every node in the Kubernetes cluster that could potentially mount the volumes provisioned by that backend. In most installations, that includes every worker node in the cluster.

For Kubernetes clusters with more than 64 nodes, you should use multiple access groups. Each access group may contain up to 64 IQNs, and each volume can belong to four access groups. With the maximum four access groups configured, any node in a cluster up to 256 nodes in size will be able to access any volume. For latest limits on volume access groups, see here.

If you're modifying the configuration from one that is using the default `trident` access group to one that uses others as well, include the ID for the `trident` access group in the list.

Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.